

VAMSHI ADEPU

Cloud Data Engineer

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SUMMARY

- **Highly skilled as a Cloud Data Engineer with over 4+ years of experience** in building and deploying **end-to-end data pipelines, cloud-native architectures, and real-time streaming solutions** using **AWS, Azure, and GCP** across finance, consulting, and technology domains.
- Maintained **ETL pipelines, data lakes, and data warehouses** with tools like **Apache Spark, AWS Glue, Airflow, Snowflake, and Azure Synapse**, enabling high-volume data processing and reporting automation.
- Delivered scalable solutions using **Python, SQL, Kafka, Redshift, BigQuery, and dbt**, with focus on **data ingestion, data modeling, machine learning, and cost-effective cloud resource optimization**.
- Experienced in **GDPR/CCPA compliance, DevOps practices**, and implementing **CI/CD pipelines** using **GitHub Actions, Jenkins, and Docker**, ensuring security, governance, and production readiness in cloud-based deployments.

PROFESSIONAL EXPERIENCE

Cloud Data Engineer

McKinsey & Company, TX | October 2023 – Current

- Designed a centralized AWS data lake using AWS Glue and Amazon S3, integrating structured/unstructured data from ERP, IoT sensors, and CRM systems to eliminate data silos and enable unified analytics.
- Engineered real-time data ingestion pipelines with AWS Kinesis Data Streams and Lambda, enabling low-latency processing of POS transactions and IoT sensor data for inventory alerts.
- Formulate predictive machine learning models in Amazon SageMaker, improving demand forecasting accuracy by 30%.
- Migrated legacy Redshift clusters to serverless RA3 nodes, optimizing performance and reducing costs.
- Enforced GDPR/CCPA compliance using AWS Lake Formation and KMS encryption, implementing column-level security for PII data across global regions.
- Automated end-to-end workflows with AWS Step Functions and CloudWatch, orchestrating ETL jobs, ML inference, and BI reporting to minimize manual intervention.
- Partnered with business analysts to design Kimball dimensional models in Redshift, creating reusable Tableau dashboards for real-time sales and logistics insights.
- Optimized AWS Glue ETL jobs by implementing partitioning and compression techniques, reducing data processing time by 40% and cutting storage costs by 20%.

Data Analyst

JP Morgan Chase, TX | February 2023 – May 2023

- Conducted financial data analysis using Python (Pandas, NumPy) and exploratory data analysis (EDA) to identify fraud patterns across transactions, delivering initial insights within 3 weeks to support risk team interventions.
- Engineered SQL-based data models in Snowflake on Azure to automate 10+ compliance reports, completing model development and deployment in 4 weeks, resulting in streamlined audit tracking.
- Designed interactive dashboards with Power BI and Azure Synapse Analytics, built in 2 weeks to enable executives to track daily KPIs across finance, operations, and compliance teams.
- Enhanced end-to-end data workflows using Azure Data Factory and Logic Apps, replacing manual processes over a span, improving workflow reliability and traceability.
- Streamlined SQL stored procedures in Azure SQL Database to handle financial datasets with 500K+ rows per run, optimizing each query within 2–3 days, reducing execution delays significantly.

Cloud Data Engineer

Dell Technologies, India | July 2020 – December 2021

- Optimized big data processing pipelines using Apache Spark, Hadoop, and PySpark, accelerating analysis of large-scale enterprise datasets.
- Deployed and managed cloud-based data warehousing solutions with Google BigQuery and AWS Redshift, reducing query execution time by 50% and improving cost efficiency.
- Enhanced ETL workflows using Apache Airflow and dbt, enhancing data ingestion reliability and minimizing manual intervention.
- Developed real-time streaming applications with Kafka, AWS Lambda, and Pub/Sub, improving customer analytics and enabling decision-making.
- Built and deployed predictive analytics models with Scikit-Learn, TensorFlow, and Keras, enhancing demand forecasting and inventory management.

- Standardized CI/CD deployment pipelines using Jenkins, GitHub Actions, and Docker, accelerating production releases while maintaining high-quality assurance.

Data Engineer

Trigent Software, India | July 2019 – June 2020

- Analyzed large datasets using SQL, Hive, and Python (Pandas, NumPy) to identify trends and generate insights for optimizing IT infrastructure and scalability strategies.
- Automated data validation processes with Python and Jupyter Notebook, increasing report accuracy by 40% and ensuring reliable data for stakeholders.
- Transformed raw data into structured formats using SQL, PostgreSQL, and Excel, enabling ad-hoc analysis and self-service reporting for business users.
- Implemented scalable data pipelines to integrate structured and unstructured data from Azure Data Lake, AWS S3, and Hadoop, streamlining cross-functional analytics.
- Developed interactive dashboards in Tableau and Power BI, empowering data-driven decision-making.

EDUCATION

University of New Haven, Connecticut

Master of Science in Business Analytics, Graduated: 2023

Lovely Professional University, Punjab, India

Bachelor of Technology in Electronics and Communication Engineering, Graduated: 2020

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, T-SQL, Java, Scala
- **Big Data & Data Processing:** Apache Spark, Hadoop, PySpark, Hive, HDFS, MapReduce, Kafka
- **Cloud Platforms:** AWS (S3, Redshift, EC2, EMR, Lambda, IAM, RDS, CloudWatch), Azure (Azure Data Factory, ADLS, Synapse, Databricks, Blob Storage, Azure Monitor), GCP (BigQuery, Dataflow, Pub/Sub, Cloud Storage)
- **ETL & Data Warehousing:** Informatica, Snowflake, Redshift, Azure Synapse, dbt (Data Build Tool), Apache Airflow
- **Data Engineering & Pipelines:** Apache Airflow, Spark Streaming, Kafka, NiFi, Batch & Streaming ETL
- **Database Technologies:** MySQL, PostgreSQL, MongoDB, Cassandra, Oracle, Azure SQL Database, Amazon Aurora
- **Machine Learning & Data Science:** TensorFlow, Keras, PyTorch, Scikit-Learn, Pandas, NumPy, Jupyter Notebook
- **Data Visualization:** Tableau, Power BI, Looker
- **Version Control & CI/CD:** Git, GitHub, GitLab, Jenkins, Bitbucket, CircleCI

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02)
- Ultimate AWS Certified Solutions Architect Associate (SAA-C03)
- Pursuing AWS Certified AI Practitioner (AIF-C01)